DO NOW — RECALL

Lesson 092 · The Three Buses: City Highways

Name: _____ Class: _____ Date: _____

1. What did PC advance to after sequential instruction at20?

2. What is MDR used for?

3. Which unit emits control signals?

4. Convert binary 10110000_2 to hexadecimal.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

21.

2. Model answer

Data/instruction transfer with memory.

3. Model answer

Control Unit.

4. Model answer

$1011\ 0000_2 = B0_{16}$.

By the end, you can…

• Address bus identifies the location the processor wants to access; typically the CPU drives address lines on a simple memory read.

• Data bus transfers the contents between memory and processor; for ordinary memory reads the bits travel memory-to-CPU.

• Control bus signals operation and coordination, including read/write; distinguish purpose and direction rather than treating all buses as one road.

Vocabulary — know these terms

address bus — Identifies location.

data bus — Transfers content bits.

control bus — Signals requests/status.

memory read — Retrieve contents of addressed location.

NEW LEARNING

Read, annotate and explain

Observe

Address bus identifies the location the processor wants to access; typically the CPU drives address lines on a simple memory read.

Explain

Data bus transfers the contents between memory and processor; for ordinary memory reads the bits travel memory-to-CPU.

Evaluate

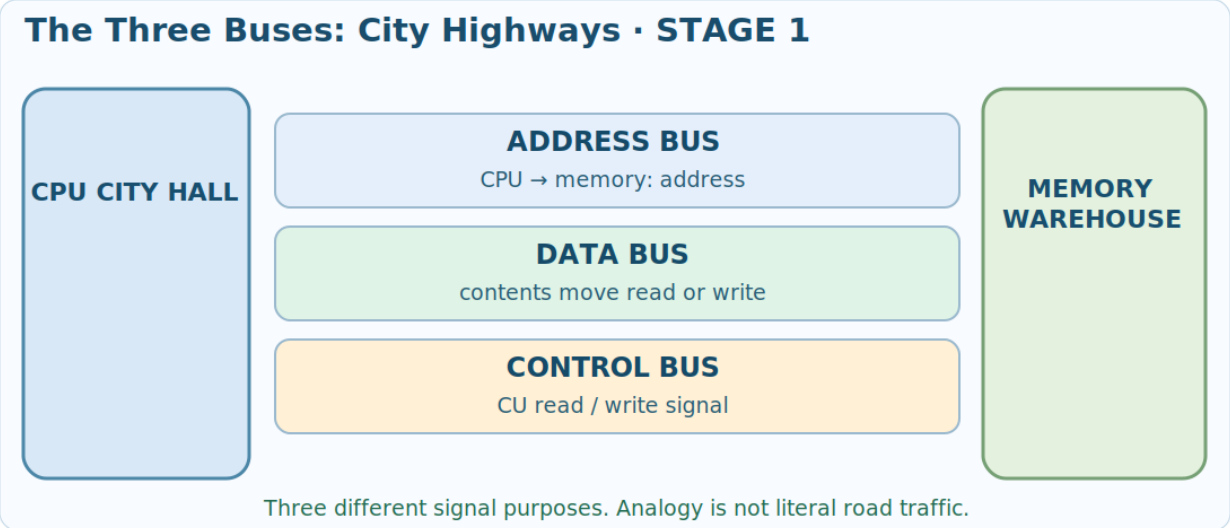
Control bus signals operation and coordination, including read/write; distinguish purpose and direction rather than treating all buses as one road.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

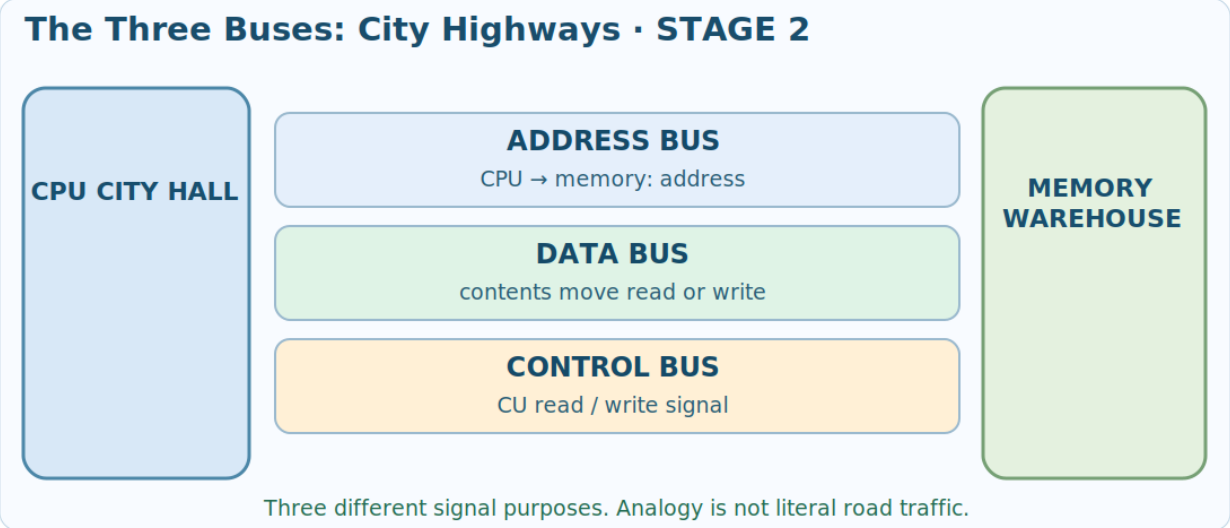
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



The Three Buses: City Highways · STAGE 1; CPU → memory: location 300

New Learning

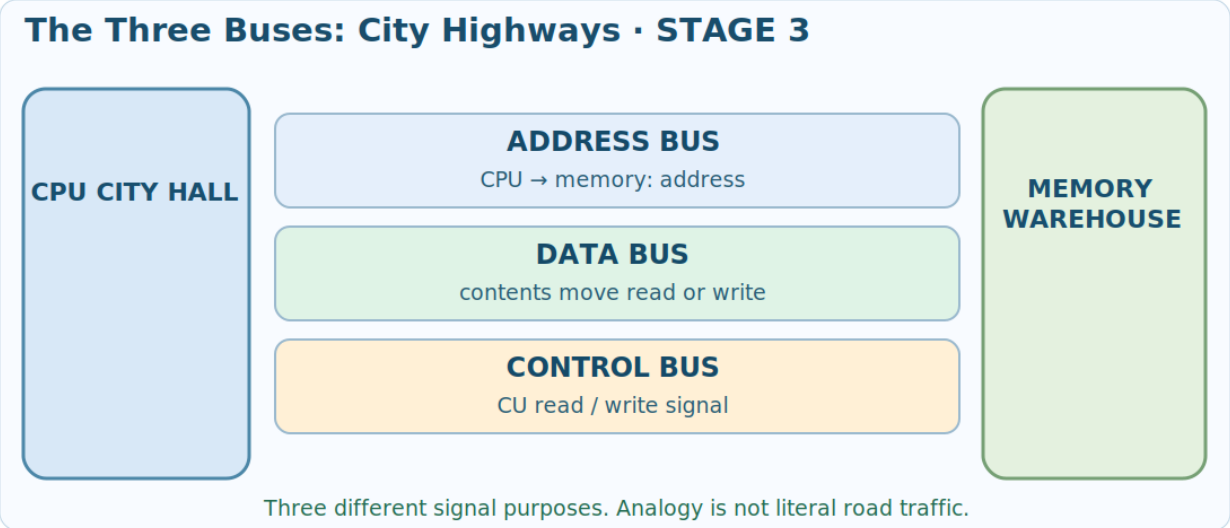


The Three Buses: City Highways · STAGE 2; CPU → memory: location 300; CU requests READ

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



The Three Buses: City Highways · STAGE 3; CPU → memory: location 300; CU requests READ; Memory → CPU: stored value 42

I DO — WORKED EXAMPLE

Follow the worked method

CPU reads memory[12]=99. Label each bus with what it carries.

Explain the mechanism

Address bus12; control bus READ; data bus returns99, placed in MDR.

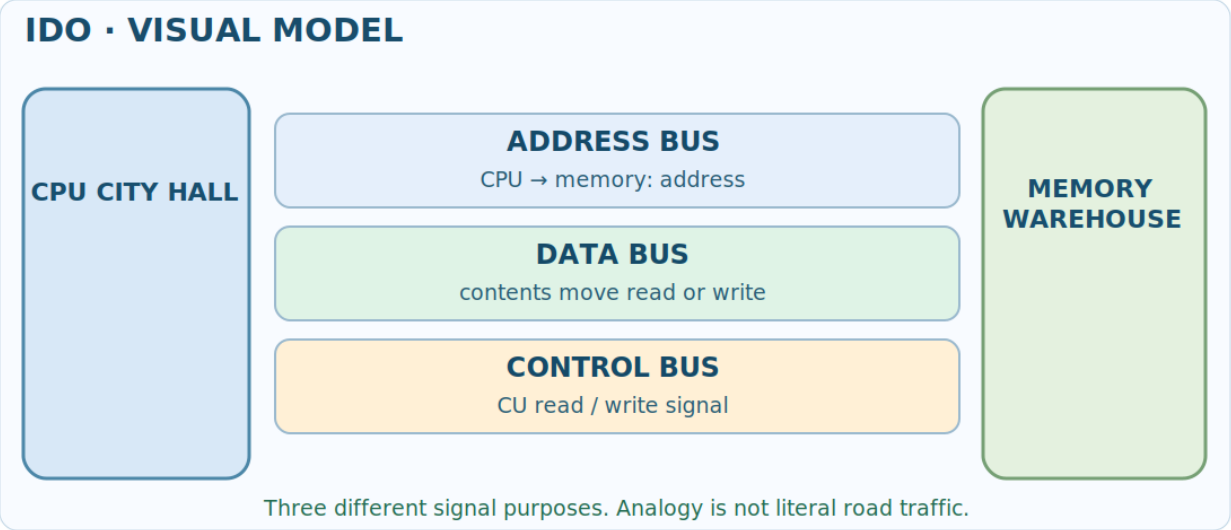
Address12 and content99 are different types of information.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

CPU reads memory[12]=99. Label each bus with what it carries.

Address bus12; control bus READ; data bus returns99, placed in MDR.



IDO · VISUAL MODEL; Address bus12; control bus READ; data bus returns99, placed in MDR.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

Guided 1 — TRY

CPU writes value25 to memory address44. State bus role for each.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

Question 1

A read from address7 returns value150. Identify address/data/control signals. [6]

Question 2

Explain role and typical direction of all three buses for a write to memory address20 of value9. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

Question 3

Evaluate why confusing addresses with contents can break CPU fetch models; use PC,MAR,MDR,CIR in example. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking.

The address bus and data bus always carry the same binary number.

Circle: Confident / Getting there / Need help

Explain what you can now do independently.

GREEN PEN — Correct and explain

Choose one error. Write the correct method and explain precisely why your first response was wrong.

WE DO — CHECK AND IMPROVE

Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

Guided 1 — CHECK

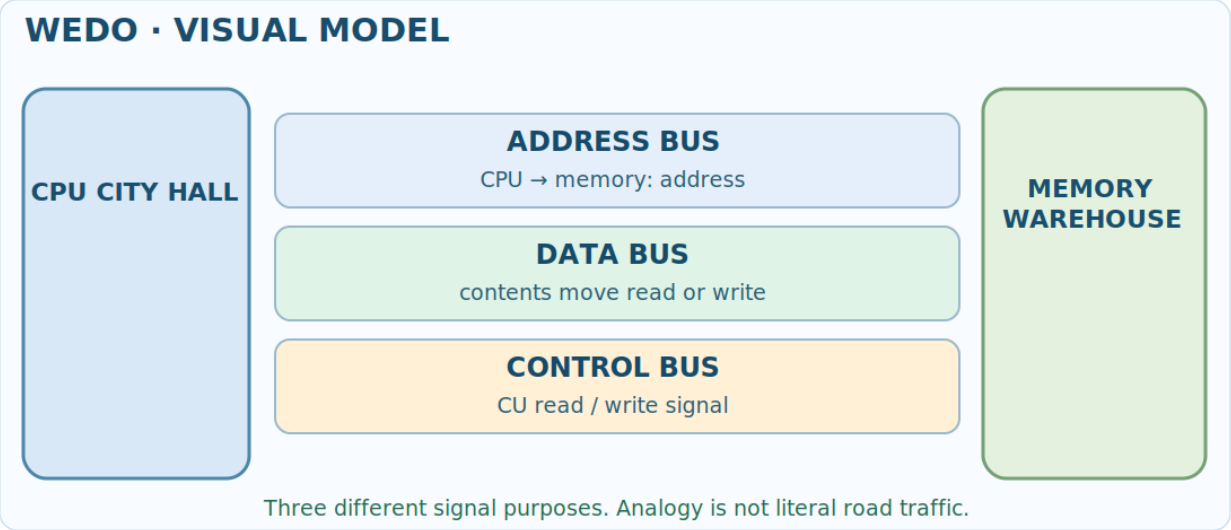
Explain the mechanism
Address bus44, data bus25 from CPU to memory, control bus WRITE.
On writing, data flows towards memory while address still identifies destination.

We Do source CHECK · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

CPU writes value25 to memory address44. State bus role for each.

Address bus44, data bus25 from CPU to memory, control bus WRITE.



WEDO · VISUAL MODEL; Address bus44, data bus25 from CPU to memory, control bus WRITE.